



Periodically intermittent discrete observation control for synchronization of the general stochastic complex network[☆]

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ABSTRACT

In this paper, the exponential synchronization of stochastic complex networks without strong connectivity is discussed. Distinguished from the previous papers, we introduce a novel periodically intermittent control based on discrete-time state observations, instead of continuous-time state observations in the control time. The coupling structure of networks is time-varying. By employing the Kirchhoff's Matrix Tree Theorem in graph theory and Tarjan's algorithm, some sufficient conditions are derived. In particular, when the novel control degenerates into the feedback control based on discrete-time state observations, it is discussed in detail as well. Furthermore, we apply our main results to study the synchronization of stochastic coupled oscillators with time-varying coupling structure. Meanwhile, two synchronization criteria are derived. Finally, a numerical test is given to validate the effectiveness of our results.

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1. Introduction

In view of the fruitful achievements of the World Wide Web (Broder et al., 2000) and scale-free topology (Ebel, Mielsch, & Bornholdt, 2002), complex networks have become an active and rich research area and their applications can be found in many fields. As a consequence, many researchers have contributed to the researches (Chen, Yu, Tan & Zhu, 2016; Meng, Chen, & Middleton, 2018; Wang, Wu, & Huang, 2015; Wu, Sun, Wen, & Wang, 2016). With the viewpoint of complex networks, a multitude of interesting and essential dynamical properties, such as synchronization, consensus and stability, have been researched (Mao, Liu, Hu, Luo, & Lu, 2014; Saravanakumar, Rajchakit, Ahn, & Karimi, 2019; Sheng, Zhang, & Zeng, 2017; Yang et al., 2017, 2019; Zhang, Yang, & Li, 2019). Synchronization, as an interesting collective property of stochastic complex networks, has become a hot topic due to its broad applications in various disciplines, and many relevant results have been reported these years (Liu, Cao, Yu, & Song, 2016; Yang, Shi, Liu, & Gao, 2011; Zhang, Li, Huang, & He, 2015). However, it is worth noting that most of the aforementioned papers considered a constant coupling structure, while in real world, the coupling structure may change over time. For

example, in reality, bidirectional associative memory neural networks ubiquitously change with time (Anbuvithya, Mathiyalagan, Sakthivel, & Prakash, 2015). Apart from that, in Chua's circuits, channels and parameters can also be time-varying (Chua, Yang, Zhong, & Wu, 1996). Therefore, it would be more significant for us to take time-varying coupling structure into consideration. Besides, due to its generality, recently synchronization analysis for stochastic complex networks with time-varying coupling structure has received great attention from researchers, and many results have been published. For example, in Zhang, Yang, Li, Zhang, and Yang (2018), time-varying coupling structure was taken into consideration by Zhang et al. when they studied stochastic exponential synchronization of delayed memristive neural networks. Besides, considering time-varying coupling structure, some techniques for global asymptotical synchronization problem of delayed fractional-order memristor-based complex-valued neural networks were presented by Yang et al. in Yang, Li, Huang, Song and Huang (2018).

Recently, many scholars have paid attention to the synchronization of complex networks (Li, Su, & Ding, 2018; Liu, Ho, Song, & Xu, 2019; Wu, Chen, & Li, 2017; Yang, Cao, Huang, Zhang & Qiu, 2018; Yang, Qiu & He, 2015). It is worth pointing out that most of the aforementioned papers held the assumptions that networks are either strongly connected or with a directed spanning tree, which is a special case in practice. In the real world, a lot of networks are developed on the basis of stochastic complex networks without strong connectivity (SCNWSCs). For example, in social networks formed of the public by social interaction, an individual stands for a node, and the degrees of

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acquaintance among each other are characterized by the weights of social digraphs. Besides, neural networks which are composed of interconnected neurons, and electric networks which are made up of the substation, transmission and distribution lines, are all established on SCNWSCs. Obviously, SCNWSCs exist everywhere in our life. Hence, SCNWSCs are more applicable and can suit the practical problems in a better way, and it has aroused considerable interest from researchers. Meanwhile, a load of results have been reported (Liu & Li, 2018; Liu, Li, & Feng, 2018). Clearly, to research the SCNWSC is of great significance. Thus in this paper, we tend to research the synchronization problem of the SCNWSC with time-varying coupling structure.

As we all know, in general, a network cannot achieve synchronization spontaneously, which indicates that we are supposed to add some external controllers on the network to realize its synchronization. In fact, these years, many control methods have been put forward. Among diverse control strategies, feedback control and intermittent control are two frequently-used types. The former one requires the control inputs to be activated continuously, while the latter one, discontinuously. In fact, control inputs in the latter one are activated during the work time and off during the rest time. Obviously, intermittent control is more economical than feedback control. Hence, it has received increasing interest from researchers, and a lot of interesting results have been reported (Chen, Zhong & Zheng, 2016; Cheng, Chen, Qiu, Lu, & Cao, 2018; Cheng et al., 2019; Ding, Wang, & Zhang, 2019; Liu & Chen, 2015; Wu, Zhu & Li, 2019; Xu et al., 2018; Xu, Zhou, & Li, 2019). We note that the controller in the aforementioned papers were observed continuously in the control time. Considering that as a rule, controller is observed at some discrete points of work time, such as $0, \tau, 2\tau, \dots$ (τ is a positive constant), therefore, it would be more general and economical on condition that a novel periodically intermittent control, which is based on discrete-time state observations in the control time, is established for the need of low cost in practical applications. As a matter of fact, Mao (2013) first came up with a discrete observation control with the aim of synchronizing the controlled systems, which merely requires state observations at discrete times $0, \tau, 2\tau, \dots$ in the control time. Due to the effectiveness and generality, it has become of high interest these days, and many scholars have contributed to the researches (Mao et al., 2014; Wu et al., 2017; You, Liu, Lu, Mao, & Qiu, 2015). Simultaneously, taking into account the high efficiency and lower cost of periodically intermittent control, in this paper, a novel control strategy called periodically intermittent discrete observation control (PIDOC), in which the states in the control time are observed on discrete time points, is firstly introduced. On the one hand, compared with regular feedback control, PIDOC is capable of reducing the cost from the perspective of economy, since it does not work during the rest time. On the other hand, PIDOC is more general and practical on account of the fact that observations for the states of the SCNWSC are at discrete time for the most part. Thus, our paper makes an attempt to research the synchronization problem of SCNWSC with time-varying coupling structure via PIDOC.

Motivated by the aforementioned discussions, in this paper, our main purpose is to research the synchronization of the SCNWSC with time-varying coupling structure. A novel PIDOC is designed in order to achieve synchronization for the first time. Secondly, by employing graph theory in combination with the Tarjan's algorithm, several sufficient conditions are derived with the aim of ensuring the synchronization. What is more, we apply the main results to stochastic coupled oscillators with time-varying coupling structure. Meanwhile, two corollaries are derived. Finally, a numerical example is shown to testify the feasibility and effectiveness of the acquired theoretical results.

The dominant contributions of this paper are as follows:

- The synchronization of the SCNWSC with time-varying coupling structure is discussed via a novel PIDOC strategy, which is a generalization of intermittent control and feedback control based on discrete-time state observations (FCBDSO).
- The coupling structure that we consider is time-varying.
- Furthermore, the topology structure is not strongly connected.
- Some novel synchronization criteria are derived through the combination of graph theory and Lyapunov method.

The rest of this paper is structured as follows. In Section 2, some preliminaries and model formulation are introduced. Next, in Section 3, we mainly discuss the synchronization of the strongly connected stochastic complex network (SCSCN) with time-varying coupling structure via PIDOC. Then in Section 4, we discuss the synchronization of the SCNWSC with time-varying coupling structure via PIDOC. Our analytical results are applied to stochastic coupled oscillators in Section 5. A numerical example is presented in Section 6. Finally, some conclusions are drawn in Section 7.

Notations. In this paper, unless otherwise specified, the superscript “T” stands for the transpose of a vector. Also, we let $\mathbb{R}$ be the set of all real numbers, $\mathbb{R}^m$ be the m -dimensional Euclidean space, and $\mathbb{Z}_+ = \{1, 2, \dots\}$, $\mathfrak{N} = \{1, 2, \dots, N\}$, $\mathbb{R}_0 = [0, +\infty)$. In addition, define $|\cdot|$ as the Euclidean norm for vectors or the trace norm of matrices. Besides, make $(\Omega, \mathcal{F}, \{\mathcal{F}^t\}_{t \geq 0}, \mathbb{P})$ be a complete probability space with a filtration $\{\mathcal{F}^t\}_{t \geq 0}$ satisfying usual conditions, and $\mathbb{E}(\cdot)$ denotes the mathematical expectation with respect to the given probability measure $\mathbb{P}$.

Let $(\mathcal{G}, A(t))$ be a weighted digraph with $N(N \geq 2)$ nodes, where $A(t) = (a_{ij}(t))_{N \times N}$ is a weighted matrix for $t \geq 0$. Furthermore, define the Laplacian matrix of digraph $(\mathcal{G}, A(t))$ as $L(t) = (p^{ij}(t))_{N \times N}$, where $p^{ij}(t) = -a^{ij}(t)$ for $i \neq j$, and $p^{ii}(t) = \sum_{j \neq i} a_{ij}(t)$. A directed path $\mathcal{P}$ in $\mathcal{G}$ is a subgraph with distinct nodes $\{i^1, i^2, \dots, i^s\}$ such that its set of arcs is $\{(i^k, i^{k+1}) : k = 1, 2, \dots, s-1\}$. If $i^s = i^1$, we call $\mathcal{P}$ a directed cycle. A digraph is called strongly connected if for any pair of distinct nodes (i, j) there exists a path from node i to node j and a path from node j to node i . Apart from that, the relation among nodes of a digraph $\mathcal{G}$ as a partial order $\prec$ is defined. For nodes $k, h \in V(\mathcal{G})$, where $V(\mathcal{G})$ represents the node set of a digraph $\mathcal{G}$, $k \prec h$ if there is a directed path from node k to node h . Then, we define a strongly connected component $\mathcal{R}$ of a digraph $\mathcal{G}$ as follows: if a subgraph $\mathcal{R}$ is strongly connected and for any node k , a subgraph consisting of the node set $V(\mathcal{R}) \cup k$ is not strongly connected, then $\mathcal{R}$ is a strongly connected component. What is more, we compress each strongly connected component of the digraph $\mathcal{G}$ into a single node. And all of them form a new digraph $\mathcal{R}$, which is considered as a condensed graph of $\mathcal{G}$. Moreover, for a condensed digraph $\mathcal{R}$, a strict partial order $\preceq$ is defined as well. Concretely, for $\mathcal{R}$, one has $\mathcal{R} \in V(\mathcal{R})$, $\mathcal{R} \prec \mathcal{R}$ if there are node $k \in \mathcal{R}$ and node $h \in \mathcal{R}$ such that $k \leq h$ in the digraph $\mathcal{G}$.

2. Preliminaries and model formulation

In this section, consider a stochastic complex network with N identical coupled nodes, which can be described as

$$dx^i(t) = f^i(x^i(t), t) + \sum_{j=1}^N a_{ij}(t)x_j(t) dt + \sum_{j=1}^N g^i(x^i(t), t)dB(t), i \in \mathfrak{N}, \quad (1)$$

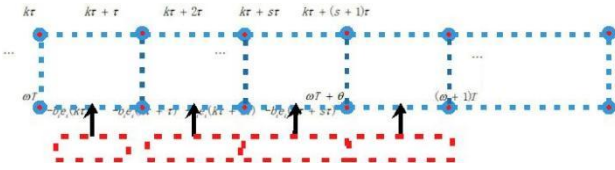


Fig. 1. Schematic diagram of the PIDOC, where $e^i(t) = y^i(t) - x^i(t)$.

where $x^i(t) = (x^{i1}(t), x^{i2}(t), \dots, x^{im}(t))^T \in \mathbb{R}^m$ represents the state vector of node i . $f, g: \mathbb{R}^m \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^m$ are continuous vector-valued functions representing the inherent nonlinear dynamics of node i , $a_{ij}(t)$ are the weights of the edges between node i and node j , $B(t)$ represents one-dimensional normal Brownian motion.

Let system (1) be the driving system, and consider the following controlled response system:

$$\begin{aligned} dy^i(t) = & f^i(y^i(t), t) + \sum_{j \in \mathcal{N}} a_{ij}(t) y^j(t) + u^i(t) dt \\ & + g^i(y^i(t), t) dB(t), i \in \mathcal{N}, \end{aligned} \quad (2)$$

where $y^i(t) = (y^{i1}(t), y^{i2}(t), \dots, y^{im}(t))^T \in \mathbb{R}^m$, $u^i(t)$ is the control input of the i th node, which is characterized as follows:

$$u_i(t) = \begin{cases} b^i(x^i(\delta(t)) - y^i(\delta(t))), & t \in [nT, nT + \theta), \\ 0, & t \in [nT + \theta, (n+1)T), \end{cases} \quad (3)$$

where $n = 0, 1, \dots$, $i \in \mathcal{N}$, $\delta(t) = [t/\tau]\tau$, $\tau > 0$ is the duration between two consecutive observations, and $[t/\tau]$ is the integer part of t/τ , $T > 0$ is the control period, θ is called the control width satisfying $0 < \theta < T$, b^i is a positive constant called control gain. Besides, we assume that $\theta/\tau \in \mathbb{Z}_+$ and $k\tau = nT(n < k, n, k \in \mathbb{Z}_+)$. Furthermore, it is supposed that there exists some positive constant a such that $|a_{ij}(t)| \leq a$, for all $i, j \in \mathcal{N}$ and for almost all $t \geq 0$.

In order to explain the control strategy in a better way, a schematic diagram is given as follows (see Fig. 1).

Moreover, let $e^i(t) = y^i(t) - x^i(t)$, $i \in \mathcal{N}$ be the error signal. Then the error system can be expressed as follows:

$$\begin{aligned} de^i(t) = & f^i(y^i(t), t) - f^i(x^i(t), t) + \sum_{j \in \mathcal{N}} a_{ij}(t) e^j(t) + u^i(t) dt \\ & + [g^i(y^i(t), t) - g^i(x^i(t), t)] dB(t), i \in \mathcal{N}. \end{aligned} \quad (4)$$

Remark 1. Recently, many scholars have studied the dynamical properties of complex networks through intermittent control in Chen, Zhong and Zheng (2016), Liu and Chen (2015) and Xu et al. (2018). For example, in Liu and Chen (2015), the control was designed as

$$u_i(t) = \begin{cases} k_i e_i(t), & t \in [nT, nT + \theta), \\ 0, & t \in [nT + \theta, (n+1)T), \end{cases}$$

where $n = 0, 1, \dots$, $i \in \mathcal{N}$. Obviously, continuous-time state observations for the state vector $e_i(t)$ were utilized in the control time ($t \in [nT, nT + \theta)$). Different from the aforementioned papers, this paper uses FCBDOS in the control time ($t \in [nT, nT + \theta)$) to study the synchronization of the SCNWS with time-varying coupling structure, which is characterized as (3). Clearly, discrete-time state observations at $0, \tau, 2\tau, \dots$ for the state vector $e^i(t)$ are needed in the control time, which is more effective and economical.

Remark 2. In this paper, the control strategy designed is PIDOC. From (3), we can obtain that when $\tau \rightarrow 0$, $b^i(x^i(\delta(t)) - y^i(\delta(t)))$

will tend to $b^i(x^i(t) - y^i(t))$ in the control time, that is to say, controller (3) is reduced to an intermittent control (Chen, Zhong & Zheng, 2016; Xu et al., 2018). Besides, when $\theta \rightarrow T$, controller (3) will degenerate into a regular FCBDOS (Mao, 2013; Mao et al., 2014; You et al., 2015). As a matter of fact, PIDOC is an extension of the intermittent control and FCBDOS. Therefore, the controller we designed is more general and less conservative.

Remark 3. In this paper, we consider the coupling as $\sum_{j=1}^N a_{ij}(t) x_j(t)$. In fact, in the case of non-linear coupling $\sum_{j=1}^N H_{ij}(x_i(t), x_j(t))$, the conclusion of this paper is also valid when the function $H_{ij}(x_i(t), x_j(t))$ satisfies the following inequality:

$$\begin{aligned} |H_{ij}(x^i(t), x^j(t)) - H_{ij}(y^i(t), y^j(t))| \\ \leq a_{ij}(|y^i(t) - x^i(t)| + |y^j(t) - x^j(t)|), \end{aligned}$$

where a_{ij} is a nonnegative constant.

To proceed, the following definition will be utilized in our main results.

Definition 1 (Mao, 1997). Drive-response systems (1) and (2) achieve exponential synchronization if there exist constants $\varpi > 0$ and $M > 0$ such that

$$\mathbb{E}|e(t)|^2 \leq M \exp\{-\varpi t\}, \quad t \geq 0,$$

where $e(t) = (e^1(t), e^2(t), \dots, e^N(t))^T$ is called the convergence rate of exponential synchronization.

3. Synchronization analysis for the SCSCN with time-varying coupling structure

In this section, synchronization of the SCSCN with time-varying coupling structure is discussed. To obtain our main results, the following assumptions are needed.

Assumption 1. For any $i, j \in \mathcal{N}$, digraph $(\mathcal{G}, A(t))$ is strongly connected, where $A(t) = (a_{ij}(t))_{N \times N}$, $c_i(t)$ is the cofactor of the i th diagonal element of Laplacian matrix of digraph $(\mathcal{G}, A(t))$, and $c_i \leq c^i(t) \leq \bar{c}^i$, $c_i(t) \leq c^i$, where $\bar{c}^i$, c^i , c^i are some positive constants.

Assumption 2. For any $i \in \mathcal{N}$, there exist positive constants ξ_i and δ_i , such that

$$\begin{aligned} |f^i(y^i(t), t) - f^i(x^i(t), t)| &\leq \xi_i |y^i(t) - x^i(t)|, \\ |g^i(y^i(t), t) - g^i(x^i(t), t)| &\leq \delta_i |y^i(t) - x^i(t)|. \end{aligned}$$

Next, for the convenience, we first introduce some notations which will be utilized throughout this paper unless otherwise specified. Set

$$\begin{aligned} b &= \max_{i \in \mathcal{N}} \{b^i\}, \bar{c} = \max_{i \in \mathcal{N}} \{\bar{c}^i\}, c = \min_{i \in \mathcal{N}} \{c^i\}, \xi = \max_{i \in \mathcal{N}} \{\xi_i\}, \\ \delta &= \max_{i \in \mathcal{N}} \{\delta_i\}, c = \max_{i \in \mathcal{N}} \{c^i\}, \bar{\delta} = \max_{i \in \mathcal{N}} \{2\xi_i + \delta_i + 2Na + \frac{c}{c}\}, \\ \beta &= \min_{i \in \mathcal{N}} \{2b^i - 2\xi_i - \delta_i - 2\lambda^1 - 2Na - \frac{k(\tau)}{1 - k(\tau)2\lambda_1} - \frac{b^2}{c}\}, \end{aligned}$$

in which λ_1 is a positive constant, and

$$k(\tau) = 2 \exp\left\{-\frac{2\bar{c}}{c} \tau\right\} \left(\frac{2\bar{c}}{c} \tau + \frac{3\tau}{\xi} + \frac{8N^2 a^2}{3} + 3\delta^2 \right).$$

Then, the following theorem can be presented.

Theorem 1. If Assumptions 1 and 2 hold, and there exists a sufficiently small τ such that

$k(\tau) < 1$, $\beta > 0$, $\theta\beta - (T - \theta)\bar{\delta} > 0$,
then drive-response systems (1) and (2) achieve exponential synchronization under controller (3).

Proof. Let function $V(e, t) = \sum_{i=1}^N c^i(t) V^i(e^i, t)$, where $V^i(e^i, t) = |e^i|^2$. Combining Assumptions 1 and 2 with the elementary inequality, one can get that

$$e^T(t)[f^i(y^i(t), t) - f^i(x^i(t), t) - b^i e^i(t)] \leq (\xi^i - b^i)|e^i(t)|^2, \quad (5)$$

$$\begin{aligned} & 2e^T(t)[b^i e^i(t) - b^i e^i(\delta(t))] \\ & \leq 2\lambda^1 |e^i(t)|^2 + \frac{b_i^2}{2\lambda_1} |e_i(\delta(t))|^2. \end{aligned} \quad (6)$$

Therefore, combining (5) with (6), when $t \in [nT, nT + \theta)$, we have

$$\begin{aligned} & \mathcal{L}V(e(t), t) \\ & \leq \sum_{i=1}^N \left[c_i(t) (2\xi^i - 2b^i + \delta^2 + 2\lambda^1 + 2a_{ij}(t)) |e_i(t)|^2 \right. \\ & \quad + \frac{b_i^2}{2\lambda_1} |e_i(t) - e_i(\delta(t))|^2 + \frac{c_i}{c} c_i(t) V(e_i(t), t) \left. \right] \quad (7) \end{aligned}$$

Now, for any $t \in [v\tau, (v+1)\tau) \cap [nT, nT + \theta)$, where $v \in \mathbb{Z}_+$, we can get that $\delta(t) = v\tau$. Therefore, we have

$$\begin{aligned} \mathbb{E}|e^i(t) - e^i(\delta(t))|^2 & \leq 3\tau \mathbb{E} \int_{\delta(t)}^t |f_i(y_i(s), s) - f_i(x_i(s), s)|^2 ds \\ & \quad + 3\tau \mathbb{E} \int_{\delta(t)}^t |a_{ij}(s)e_j(s) + b^i e^i(\delta(s))|^2 ds \\ & \quad + 3\tau \mathbb{E} \int_{\delta(t)}^t |g^i(y^i(s), s) - g^i(x^i(s), s)|^2 ds. \end{aligned}$$

In view of Assumptions 1 and 2, one can derive that

$$\begin{aligned} & \mathbb{E}|e^i(t) - e^i(\delta(t))|^2 \\ & \leq \int_{\delta(t)}^t 3\tau \xi^i + 3\delta^2 \mathbb{E}|e_i(s)|^2 ds \\ & \quad + 6\tau \int_{\delta(t)}^t b^i \mathbb{E}|e_i(\delta(s))|^2 + N \sum_{j=1}^N a_{ij}^2(s) \mathbb{E}|e_j(s)|^2 ds \\ & \leq \int_{\delta(t)}^t 3\tau \xi^i + 3\delta^2 \mathbb{E}|e_i(s)|^2 ds + 6\tau \sum_{j=1}^N b_i^2 \mathbb{E}|e_j(\delta(t))|^2 \\ & \quad + 6\tau N \sum_{j=1}^N a_{ij}^2(s) \mathbb{E}|e_j(s)|^2 ds. \end{aligned} \quad (8)$$

Consequently, according to Assumption 1 and (8), it yields

$$\begin{aligned} & c^i(t) \mathbb{E}|e^i(t) - e^i(\delta(t))|^2 \\ & \leq \int_{\delta(t)}^t \left[c^i(s) (3\tau \xi^i + 2Na_{ij}(s)) \right. \\ & \quad + 3\delta_i^2 \mathbb{E}|e_i(s)|^2 + 6\tau \sum_{j=1}^N c^i(t) b_i^2 \mathbb{E}|e_j(\delta(t))|^2 \\ & \quad \left. + 6\tau Na_{ij}^2(s) \mathbb{E}|e_j(s)|^2 \right] ds. \end{aligned}$$

Considering that digraph $(\mathcal{G}, A(t))$ is strongly connected, according to Theorem 2.2 in Li and Shuai (2010), it is clear that

$$\begin{aligned} & \sum_{i=1}^N \sum_{j=1}^N c^i(t) a_{ij}(t) [|e^j(t)|^2 - |e^i(t)|^2] \\ & = \sum_{\theta \in \mathcal{C}} W(\theta) [|e^r(t)|^2 - |e^s(t)|^2], \end{aligned}$$

where C_θ denotes the directed cycle of θ . From the fact that

$$\sum_{(s,r) \in E(C_\theta)} [|e^s|^2 - |e^r|^2] = 0, \text{ one has}$$

$$\sum_{i=1}^N c^i(t) a_{ij}(t) \mathbb{E}[|e^j(t)|^2 - |e^i(t)|^2] = 0.$$

As a consequence, we have

$$\begin{aligned} & \sum_{i=1}^N c^i(t) \mathbb{E}|e^i(t) - e^i(\delta(t))|^2 \\ & \leq \int_{\delta(t)}^t \left[\sum_{i=1}^N c^i(s) (3\tau \xi^i + 2Na_{ij}(s)) \right. \\ & \quad \left. + 3\delta_i^2 \mathbb{E}|e_i(s)|^2 + 6\tau \sum_{j=1}^N c^i(t) b_i^2 \mathbb{E}|e_j(\delta(t))|^2 \right] ds. \end{aligned}$$

Based on elementary inequality, one can get

$$\begin{aligned} & \sum_{i=1}^N c^i(t) \mathbb{E}|e^i(t) - e^i(\delta(t))|^2 \\ & \leq 6\tau \sum_{i=1}^N \bar{c} b^2 + \frac{2\bar{c}^2}{c^2} \tau \sum_{i=1}^N c^i(t) \mathbb{E}|e^i(\delta(t))|^2 \\ & \quad + \int_{\delta(t)}^t \frac{2\bar{c}}{c} \sum_{i=1}^N c_i(s) \mathbb{E}|e_i(s) - e_i(\delta(s))|^2 ds \\ & \leq 2 \sum_{i=1}^N 6\tau \bar{c} b^2 + \frac{2\bar{c}^2}{c^2} \tau \sum_{i=1}^N c_i(t) \\ & \quad \cdot (\mathbb{E}|e^i(t)|^2 + \mathbb{E}|e^i(t) - e^i(\delta(t))|^2) \exp \frac{2\bar{c}}{c} \tau \sum_{i=1}^N c_i, \end{aligned}$$

where $\sum_{i=1}^N c_i = 3\tau \sum_{i=1}^N \xi^i + 2N^2 a^2 + 3\delta^2$. Meanwhile, on account of the fact that $k(\tau) < 1$, we can obtain that

$$c^i(t) \mathbb{E}|e^i(t) - e^i(\delta(t))|^2 \leq \frac{k(\tau)}{1 - k(\tau)} c^i(t) \mathbb{E}|e^i(t)|^2. \quad (9)$$

Consequently, combining (7) with (9), we can verify that

$$\begin{aligned} & \mathbb{E} \mathcal{L}V(e(t), t) \\ & \leq \sum_{i=1}^N c_i(t) (2\xi^i - 2b^i + \delta^2 + 2\lambda^1 + 2a_{ij}(t)) \\ & \quad + \frac{k(\tau)}{1 - k(\tau)} \frac{b^2}{2\lambda_1} \mathbb{E}|e^i(t)|^2 + \frac{c}{c} \sum_{i=1}^N c_i(t) \mathbb{E}V_i(e_i(t), t). \end{aligned}$$

Therefore, according to definition of β , it can be concluded that

$$\mathbb{E} \mathcal{L}V(e(t), t) \leq -\beta \mathbb{E}V(e(t), t).$$

Then, the following inequality can be derived

$$\mathbb{E}V(e(t), t) \leq \mathbb{E}V(e(\omega T), \omega T) \exp\{-\beta(t - \omega T)\}. \quad (10)$$

Similarly, when $t \in [nT + \theta, (n+1)T)$, according to Assumptions 1 and 2, it can be derived that

$$\begin{aligned} & \mathcal{L}V(e(t), t) \\ & \leq \sum_{i=1}^N c^i(t) (2\bar{\xi}^i + \delta^i)^2 + 2 \sum_{i=1}^N a_{ij}(t) |e_i(t)|^2 \\ & \quad + c^i(t) V(e^i, t) + c^i(t) a^{ij}(t) [|e^i(t)|^2 - |e_i(t)|^2]. \end{aligned}$$

Consequently, combined with (3), one has

$$\mathcal{L}V(e(t), t) \leq \bar{\delta}V(e(t), t).$$

As a consequence, we can obtain that

$$\mathbb{E}V(e(t), t) \leq \mathbb{E}V(e(\omega T + \theta), \omega T + \theta) \exp\{\bar{\delta}(t - \theta - \omega T)\}.$$

Hence, it can be derived that when $0 \leq t < \theta$,

$$\mathbb{E}V(e(t), t) \leq V(e(0), 0) \exp\{-\beta t\},$$

$$\mathbb{E}V(e(\theta), \theta) \leq V(e(0), 0) \exp\{-\beta\theta\},$$

when $\theta \leq t < T$,

$$\begin{aligned} \mathbb{E}V(e(t), t) & \leq \mathbb{E}V(e(\theta), \theta) \exp\{\bar{\delta}(t - \theta)\} \\ & \leq V(e(0), 0) \exp\{-\beta\theta + \bar{\delta}(t - \theta)\}, \end{aligned}$$

$$\begin{aligned} \mathbb{E}V(e(T), T) & \leq \mathbb{E}V(e(\theta), \theta) \exp\{\bar{\delta}(T - \theta)\} \\ & \leq V(e(0), 0) \exp\{-\beta\theta + \bar{\delta}(T - \theta)\}. \end{aligned}$$

Consequently, based on the above process, we can derive that when $T \leq t < T + \theta$,

$$\begin{aligned} \mathbb{E}V(e(t), t) & \leq \mathbb{E}V(e(T), T) \exp\{\beta(t - T)\} \\ & \leq V(e(0), 0) \exp\{-\beta\theta - \beta(t - T) + \bar{\delta}(T - \theta)\}, \end{aligned}$$

$$\begin{aligned} \mathbb{E}V(e(T + \theta), T + \theta) & \leq V(e(0), 0) \exp\{-2\beta\theta + \bar{\delta}(T - \theta)\}, \\ \text{when } T + \theta \leq t < 2T, \end{aligned}$$

$$\begin{aligned} \mathbb{E}V(e(t), t) & \leq V(e(0), 0) \exp\{-2\beta\theta + \bar{\delta}(T - \theta) + \bar{\delta}(t - T - \theta)\}, \\ \mathbb{E}V(e(2T), 2T) & \leq V(e(0), 0) \exp\{-2\beta\theta + 2\bar{\delta}(T - \theta)\}. \end{aligned}$$

According to the induction, it can be derived that when $nT \leq t < nT + \theta$, that is $(t - \theta)/T < n \leq t/T$, we have

$$\begin{aligned} \mathbb{E}V(e(t), t) & \leq \mathbb{E}V(e(nT), nT) \exp\{-\beta(t - nT)\} \\ & \leq V(e(0), 0) \exp\{-n\beta\theta + n\bar{\delta}(T - \theta)\} \\ & \leq V(e(0), 0) \exp\left\{-\frac{\theta\beta - (T - \theta)\bar{\delta}}{T}(t - \theta)\right\}, \\ \mathbb{E}V(e(t), t) & \leq V(e(0), 0) \exp\{(-n - 1)\beta\theta + (n + 1)\bar{\delta}(T - \theta)\} \\ & \leq V(e(0), 0) \exp\left\{-\frac{\theta\beta - (T - \theta)\bar{\delta}}{T}t\right\} \\ & \leq V(e(0), 0) \exp\left\{-\frac{\theta\beta - (T - \theta)\bar{\delta}}{T}(t - \theta)\right\}. \end{aligned}$$

Overall, for any $t > 0$, we can obtain that

$$\mathbb{E}V(e(t), t) \leq V(e(0), 0) \exp\left\{-\frac{\theta\beta - (T - \theta)\bar{\delta}}{T}(t - \theta)\right\}.$$

Moreover, consider that $V(e(t), t) = \sum_{i=1}^N c^i(t) |e_i(t)|^2 \geq c |e(t)|^2$. Hence, it can be derived that

$$\mathbb{E}|e(t)|^2 \leq \frac{V(e(0), 0)}{c} \exp\left\{-\frac{\theta\beta - (T - \theta)\bar{\delta}}{T}(t - \theta)\right\}.$$

What is more, considering that $\theta\beta - (T - \theta)\bar{\delta} > 0$, and according to Definition 1, we can derive that systems (1) and (2) achieve exponential synchronization. The proof is completed. $\square$

Remark 4. Theorem 1 provides a synchronization criterion to guarantee the synchronization of SCSCN with time-varying coupling structure. As a matter of fact, from Theorem 1, we can derive that when the time-varying coupling structure is reduced to a constant one, the results in our paper still hold. Besides, in Liu et al. (2018), it was required that $a_{ij}(t)$ must be monotone decreasing, which is a relatively special case in practical problems. While in this paper, we relax the condition that it can merely be bounded so as to be more general.

Next, we will consider another case that when $\theta \rightarrow T$, the PIDOC will degenerate into a regular FCBDOS, which is characterized as follows:

$$u^i(t) = b^i(x^i(\delta(t)) - y^i(\delta(t))), \quad i \in \mathcal{N}. \quad (11)$$

In order to study the synchronization problem of systems (1) and (2) under controller (11), we present the following definition about synchronization in mean square.

Definition 2 (Yang, Zhou et al., 2015). Systems (1) and (2) achieve synchronization in mean square if the error system (4) is stable in mean square, that is, for any initial value $e(0) \in \mathbb{R}^{mN}$, it is satisfied that

$$\lim_{t \rightarrow \infty} \mathbb{E} \int_0^t |e(s)|^2 ds < \infty.$$

Then, two corollaries can be derived as follows.

Corollary 1. If Assumptions 1 and 2 hold, and there exists a sufficiently small τ such that $k(\tau) < 1$, $\beta > 0$, then systems (1) and (2) achieve synchronization in mean square under controller (11).

Corollary 2. If Assumptions 1 and 2 hold, and there exists a sufficiently small τ such that $k(\tau) < 1$, $\beta > 0$, then systems (1) and (2) achieve exponential synchronization under controller (11).

Remark 5. When $\theta \rightarrow T$, PIDOC (3) will degenerate into a regular FCBDOS (11) and this control has been considered by many scholars. For example, Mao et al. used this control to study the stability of stochastic systems without coupling structure in Mao (2013). In Wu et al. (2017), we studied the stability of complex networks with constant or time-varying coupling structure, and the topological structure requires strong connectivity. In this paper, we study the synchronization of stochastic complex networks without strong connectivity by FCBDOS (11), which is a generalization of previous literature.

4. Synchronization analysis for the SCNWSC with time-varying coupling structure

In the previous section, synchronization of the SCSCN was studied. Now, we will change our focus to synchronization of the SCNWSC. In fact, for the SCNWSC, the well-known Tarjan's algorithm (Tarjan, 1972) can be employed to find all strongly connected components of a digraph. If digraph $(\mathcal{G}, A(t))$ is not strongly connected, we introduce a hierarchical method for it in detail (Liu & Li, 2018; Liu et al., 2018).

1. At first, for a digraph $(\mathcal{G}, A(t))$, we are supposed to compress each strongly connected component into a node, then we get a condensed digraph $\mathcal{A}$.
2. Then, for the condensed digraph $\mathcal{A}$, we place all nodes with zero in-degree in the first layer.
3. Furthermore, we say that node $\mathcal{D}$ belongs to the k th ($k \geq 2$) layer if and only if it satisfies the following conditions:

- There is a node $\mathcal{C}$ belonging to the $(k - 1)$ th layer and satisfying that $\mathcal{C} \prec \mathcal{D}$.

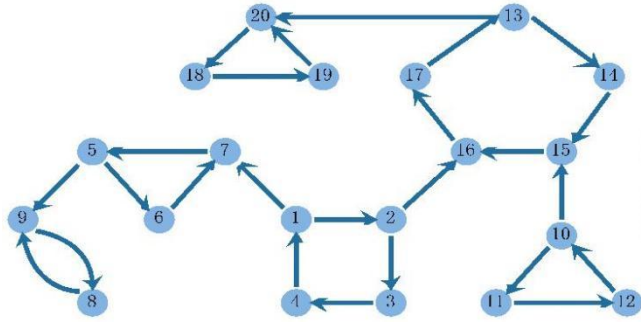


Fig. 2. A digraph $(G, A(t))$ with 20 nodes which is not strongly connected.

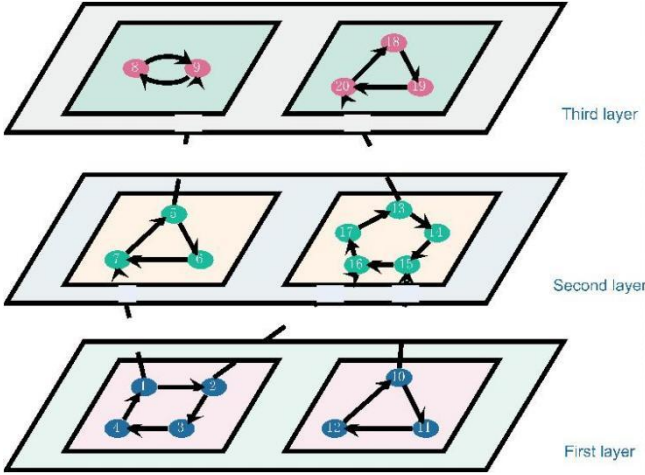


Fig. 3. The condensed digraph $\mathcal{A}$ with three layers, each group is a strongly connected component.

- If there exists a node B such that $B < D$, then the serial number of the layer that node B belongs to will not exceed $k - 1$.

4. Finally, the layered work will be completed when all nodes in condensed digraph $\mathcal{A}$ have corresponded to a certain layer.

For better comprehension of the hierarchical method for the SCNWSC, the following diagrams are presented. Here, Fig. 2 presents an SCNWSC with 20 nodes, and Fig. 3 shows a hierarchical network, in which the green layer represents the first layer, the blue layer represents the second one, and the gray layer represents the third one.

Then, in order to obtain our main results, the following hypothesis is required.

Assumption 3. For any $i \in \mathfrak{N}$, $\lim_{t \rightarrow \infty} f_i(x_i, t) = f_i(x_i)$, $\lim_{t \rightarrow \infty} g_i(x_i, t) = g_i(x_i)$, $\lim_{t \rightarrow \infty} a_i(t) = a_i$. And there exist positive constants $\bar{\xi}_i$ and $\hat{\delta}_i$ such that

$$|\tilde{f}(y^i) - \tilde{f}(x^i)| \leq \bar{\xi}_i |y^i - x^i|, \quad |\tilde{g}(y^i) - \tilde{g}(x^i)| \leq \hat{\delta}_i |y^i - x^i|.$$

For the convenience, we make some notations. Set

$$b = \max_{i \in \mathfrak{N}} \{b^i\}, \quad \bar{\xi} = \max_{i \in \mathfrak{N}} \{\bar{\xi}_i\}, \quad \hat{\delta} = \max_{i \in \mathfrak{N}} \{\hat{\delta}_i\},$$

$$\bar{\epsilon} = 2\bar{\xi} + \hat{\delta}^2 + 4N\hat{a}, \quad \hat{a} = \max_{i,j \in \mathfrak{N}} \{a^{ij}\},$$

$$\epsilon = -\frac{2\bar{\xi}}{\lambda^1} - 2b + 2\lambda^1 + \hat{\delta}^2 + 4N\hat{a} + \frac{b^2}{2\lambda^1} \frac{k_1(\tau)}{1 - k_1(\tau)},$$

in which λ_1 is a positive constant, and

$$k_1(\tau) = 2 \left(6\tau^2 b^2 + 4\tau \left(3\tau \bar{\xi}^2 + 4Na^2 + \hat{\delta}^2 \right) \cdot \exp \left(4\tau \left(3\tau \bar{\xi}^2 + 4Na^2 + \hat{\delta}^2 \right) \right) \right).$$

Besides, we introduce some explanations of notations.

$D(T_{ik})$: the node set of the k th strongly connected component of the i th layer of digraph $(G, (a_{ij}(t))_{N \times N})$.

$\cup_k D(T)$: the set of nodes composed of all the strongly connected component of the i th layer of digraph $(G, (a_{ij}(t))_{N \times N})$.

As a consequence, based on the hierarchical method and graph theory, the synchronization criterion about the SCNWSC with time-varying coupling structure is proposed as follows.

Theorem 2. When $i \in \cup_k D(T^{1k})$, all the conditions of Theorem 1 are satisfied. When $i \in \mathfrak{N} \setminus \cup_k D(T^{1k})$, if Assumption 3 holds, and there exists a sufficiently small τ such that

$$k_1(\tau) < 1, \quad \epsilon > 0, \quad \theta \epsilon - (T - \theta)\bar{\epsilon} > 0, \quad (12)$$

then drive-response systems (1) and (2) achieve exponential synchronization under controller (3).

Proof. We divide the proof into two parts. Firstly, we consider the first layer. Based on the hierarchical method, on account of the fact that the first layer is not influenced by other layers, we can get the error system restricted to the first layer as follows: when $t \in [nT, nT + \theta)$,

$$\begin{aligned} de_i(t) = & f^i(y^i(t), t) - f^i(x^i(t), t) \\ & + \left[\sum_{j \in \mathfrak{N}(T_{1k})} a^{ij}(t)e^j(t) - b^i e^i(\delta(t)) \right] dt \\ & + [g^i(y^i(t), t) - g^i(x^i(t), t)] dB(t), \quad i \in D(T_{1k}), \end{aligned}$$

when $t \in [nT + \theta, (n+1)T)$,

$$\begin{aligned} de_i(t) = & f^i(y^i(t), t) - f^i(x^i(t), t) + \sum_{j \in \mathfrak{N}(T_{1k})} a_{ij}(t)e_j(t) dt \\ & + [g^i(y^i(t), t) - g^i(x^i(t), t)] dB(t), \quad i \in D(T_{1k}). \end{aligned}$$

Then, for the first layer T_{1k} , let $V_{1k}^k(e^{1k}, t) = |e^i|^{2k}$. Here, e^{1k} denotes the state of the k th strongly connected component of the first layer, $c_i^{(1k)}(t)$ denotes the cofactor of the i th diagonal element of Laplacian matrix of digraph $(G, (a_{ij}(t))_{i,j \in D(T_{1k})})$. Considering that T is a strongly connected component of the first layer, in virtue of the proof of Theorem 1, we know that networks on the first layer achieve exponential synchronization.

Next, for the second layer T_{2k} , firstly we describe the error system restricted on the second layer as follows: when $t \in [nT, nT + \theta)$,

$$\begin{aligned} de_i(t) = & f^i(y^i(t), t) - f^i(x^i(t), t) + \sum_{j \in \mathfrak{N}(T_{2k})} a_{ij}(t)e_j(t) dt \\ & + [g^i(y^i(t), t) - g^i(x^i(t), t)] dB(t), \quad i \in D(T_{2k}), \end{aligned}$$

when $t \in [nT + \theta, (n+1)T)$,

$$\begin{aligned} de_i(t) = & f^i(y^i(t), t) - f^i(x^i(t), t) \\ & + \left[\sum_{j \in \mathfrak{N}(T_{2k})} a_{ij}(t)e_j(t) + \sum_{j \in \mathfrak{N}(T_{1k})} a_{ij}(t)e_j(t) \right] dt \\ & + [g^i(y^i(t), t) - g^i(x^i(t), t)] dB(t), \quad i \in D(T_{2k}). \end{aligned}$$

By employing the theory of asymptotically autonomous systems in [Thieme \(1994\)](#) and [Assumption 3](#), we can get the error system as follows: when $t \in [nT, nT + \theta]$,

$$de^i(t) = \tilde{f}(y^i) - \tilde{f}(x^i) + \sum_{j \in \mathcal{R}_{2k}^{(2k)}} a_{ij} e^j(t) - b^i e^i(\delta(t)) \, dt + [\tilde{g}^i(y^i) - \tilde{g}^i(x^i)] dB(t), \quad i \in D(T_{2k}),$$

when $t \in [nT + \theta, (n+1)T]$,

$$de^i(t) = \tilde{f}(y^i) - \tilde{f}(x^i) + \sum_{j \in \mathcal{R}_{2k}^{(2k)}} a_{ij} e^j(t) \, dt + [\tilde{g}^i(y^i) - \tilde{g}^i(x^i)] dB(t), \quad i \in D(T_{2k}).$$

Next, we construct the Lyapunov function $V_{2k}(e^{2k}, t) = \sum_{i=1}^N c_i^{(2k)} V_i(e^i, t)$, where $V_i(e^i, t) = |e^i|^2$. Here e denotes the state of the k th strongly connected component of the second layer, $c_i^{(2k)}$ denotes the cofactor of the i th diagonal element of Laplacian matrix of digraph $(\mathcal{G}, (a_{ij})_{i,j \in D(T_{2k})})$. Consider that T_{2k} is a strongly connected component of the second layer, according to Proposition 2.1 in [Li and Shuai \(2010\)](#), we have $c_i^{(2k)} > 0$.

Meanwhile, when $t \in [nT, nT + \theta]$,

$$\mathcal{L}V_i(e_i(t), t) = 2e_i^T(t) [\tilde{f}(y^i(t)) - \tilde{f}(x^i(t)) + \sum_{j \in \mathcal{R}_{2k}^{(2k)}} a_{ij} e^j(t) - b^i e^i(\delta(t)) + |\tilde{g}^i(y^i(t)) - \tilde{g}^i(x^i(t))|].$$

In view of [Assumption 3](#), it can be derived that

$$\begin{aligned} \mathcal{L}V_i(e_i(t), t) &\leq (2\xi_i - 2b_i) |e_i(t)|^2 + 2\lambda_1 |e_i(t)|^2 + \delta_i^2 |e_i(t)|^2 \\ &\quad + \frac{b_i^2}{2\lambda_1} |e_i(t) - e_i(\delta(t))|^2 + 2 \sum_{j \in \mathcal{R}_{2k}^{(2k)}} a_{ij} |e_i(t)|^2 \\ &\quad + \sum_{j \in \mathcal{R}_{2k}^{(2k)}} a_{ij} |e^j(t)|^2 - |e_i(t)|^2. \end{aligned}$$

Therefore, we have

$$\begin{aligned} \mathcal{L}V_{2k}(e^{2k}(t), t) &\leq \sum_{i \in \mathcal{R}_{2k}^{(2k)}} c_i^{(2k)} (2\xi_i - 2b_i) |e^i(t)|^2 + 2\lambda_1 |e^i(t)|^2 \\ &\quad + \delta_i^2 |e^i(t)|^2 + 2 \sum_{j \in D(T_{2k})} a_{ij} |e^i(t)|^2 \\ &\quad + \sum_{i \in \mathcal{R}_{2k}^{(2k)}} c_i^{(2k)} \frac{b_i^2}{2\lambda_1} |e^i(t) - e^i(\delta(t))|^2 \\ &\quad + \sum_{i \in \mathcal{R}_{2k}^{(2k)}} a_{ij} c_i^{(2k)} |e^i(t)|^2 - |e^i(t)|^2. \end{aligned}$$

According to Theorem 2.2 in [Li and Shuai \(2010\)](#), we know that

$$\sum_{i \in D(T_{2k})} \sum_{j \in \mathcal{R}_{2k}^{(2k)}} a_{ij} c_i^{(2k)} |e^i(t)|^2 - |e^j(t)|^2 = 0. \quad (13)$$

With the similar analysis to (8), according to [Assumptions 1](#) and [2](#), one can get that

$$\begin{aligned} &\sum_{i \in \mathcal{R}_{2k}^{(2k)}} c_i^{(2k)} \mathbb{E} |e^i(t) - e^i(\delta(t))|^2 \\ &\leq \int_0^t \left[3\tau \xi^2 + 2 \hat{a}^2 + \delta^2 \sum_{i \in \mathcal{R}_{2k}^{(2k)}} c_i^{(2k)} \mathbb{E} |e^i(s)|^2 \right. \\ &\quad \left. + 6\tau^2 \sum_{i \in \mathcal{R}_{2k}^{(2k)}} c_i^{(2k)} \frac{b_i^2}{2\lambda_1} \mathbb{E} |e^i(s) - e^i(\delta(s))|^2 \right. \\ &\quad \left. + 12\tau \sum_{i \in D(T_{2k})} \sum_{j \in D(T_{2k})} c_i^{(2k)} a_{ij} \mathbb{E} |e^j(s)|^2 - |e^i(s)|^2 \right] ds \end{aligned}$$

$$\leq 2 \sum_{i \in \mathcal{R}_{2k}^{(2k)}} c_i^{(2k)} \left[6\tau^2 b^2 + 4\tau \sum_{j \in \mathcal{R}_{2k}^{(2k)}} \xi_j^{-2} + 2 \hat{a}^2 + \delta^2 \right] \mathbb{E} |e^i(t)|^2 + \mathbb{E} |e^i(t) - e^i(\delta(t))|^2 \exp \left[4\tau \sum_{j \in \mathcal{R}_{2k}^{(2k)}} \xi_j^2 + 2 \hat{a}^2 + \delta^2 \right].$$

In view of the fact that $k_1(\tau) \leq 1$, we can derive that

$$\begin{aligned} &\sum_{i \in \mathcal{R}_{2k}^{(2k)}} c_i^{(2k)} \mathbb{E} |e(t) - e(\delta(t))|^2 \leq \pi \tau \sum_{i \in \mathcal{R}_{2k}^{(2k)}} c_i^{(2k)} \mathbb{E} |e(t)|^2 \\ &\text{where } \pi(\tau) = \frac{k_1(\tau)}{1 - k_1(\tau)}. \text{ So, we can conclude that} \\ &\mathbb{E} \mathcal{L}V_{2k}(e_i^{2k}(t), t) \leq \sum_{i \in \mathcal{R}_{2k}^{(2k)}} c_i^{(2k)} [2\xi_i - 2b_i + 2\lambda_1 + \delta_i^2] \\ &\quad + 2 \sum_{j \in \mathcal{R}_{2k}^{(2k)}} a_{ij} + \frac{b_i^2}{2\lambda_1} \frac{k_1(\tau)}{1 - k_1(\tau)} \mathbb{E} |e_i(t)|^2 \\ &\leq -\epsilon \mathbb{E} V_{2k}(e_i^{2k}(t), t). \end{aligned}$$

Similarly, one can get that

$$\begin{aligned} \mathbb{E} \mathcal{L}V_{2k}(e_i^{2k}(t), t) &\leq \sum_{i \in \mathcal{R}_{2k}^{(2k)}} c_i^{(2k)} [2e_i^T(t) f(y^i(t)) - f(x^i(t))] \\ &\quad + \sum_{j \in \mathcal{R}_{2k}^{(2k)}} a_{ij} e^j(t) + |\tilde{g}^i(y^i(t)) - \tilde{g}^i(x^i(t))|^2 \\ &\leq \sum_{i \in \mathcal{R}_{2k}^{(2k)}} c_i^{(2k)} [2\xi_i + \delta_i^2 + 2 \sum_{j \in \mathcal{R}_{2k}^{(2k)}} a_{ij} \mathbb{E} |e_i(t)|^2] \\ &= \epsilon \mathbb{E} V_{2k}(e_i^{2k}(t), t). \end{aligned}$$

As a result, it can be obtained that

$$\mathbb{E} V_{2k}(e_i^{2k}(t), t) < \begin{cases} -\epsilon \mathbb{E} V_{2k}(e_i^{2k}(t), t), & t \in [nT, nT + \theta], \\ \epsilon \mathbb{E} V_{2k}(e_i^{2k}(t), t), & t \in [nT + \theta, (n+1)T]. \end{cases}$$

Therefore, according to the similar proof of [Theorem 1](#), combined with (12), we can verify that systems (1) and (2) on the second layer achieve exponential synchronization. Repeating above-mentioned process, we can verify that systems (1) and (2) restricted on every strongly connected component of each layer achieve exponential synchronization. Therefore, systems (1) and (2) achieve exponential synchronization. The proof is completed. $\square$

Remark 6. With the purpose of analyzing the exponential synchronization for the SCNWSC with time-varying coupling structure, [Theorem 2](#) is presented. In the paper, based on Tarjan's algorithm, we divide an SCNWSC into several strongly connected components, while for each of them we can obtain that the cofactor of every diagonal element of the Laplacian matrix of the corresponding digraph is positive, which plays an essential role in the analysis of exponential synchronization.

Remark 7. Recently, many scholars studied the synchronization of stochastic complex networks. For example, in [Wu et al. \(2017\)](#), we considered the synchronization issue of stochastic coupled systems on networks with strong connectivity. Besides, in [Guo, Wu, Xiao, and Zhang \(2018\)](#), Guo et al. studied the synchronization of stochastic coupled systems on strongly connected networks. It is worth noting that the aforementioned papers held the assumption that the digraph is strongly connected, while in our paper, we relax the condition that the digraph can be not strongly connected. In particular, when the SCNWSC degenerates into the SCSCN, the results in our paper still hold, which implies the generality of our paper.

5. An application to stochastic coupled oscillators with time-varying coupling structure

In this section, by employing the main results, we discuss the synchronization of time-varying stochastic coupled oscillators (TSCOs), which have been researched by many scholars (Guo et al., 2018; Wang, Zhang, & Su, 2019; Wu, Li & Li, 2019).

In this paper, we consider the following second-order oscillators with stochastic perturbation

$$\ddot{x}^i(t) + \Phi \dot{x}^i(t) + \rho x^i(t) = \gamma x^i(t) \dot{B}(t), i \in \mathfrak{N}.$$

where Φ , ρ and γ are positive constants.

Set $\tilde{x}^i(t) = \dot{x}^i(t) + \mu x^i(t)$, then the above system can be expressed as follows:

$$\begin{aligned} dx^i(t) &= [\tilde{x}^i(t) - \mu x^i(t)] dt, \\ d\tilde{x}^i(t) &= [(\mu - \Phi)(\tilde{x}^i(t) - \mu x^i(t)) - \rho x^i(t)] dt \\ &\quad + \gamma x^i(t) dB(t), i \in \mathfrak{N}. \end{aligned}$$

Next, we take time-varying coupling structure into account. Consequently, the following TSCOs on network with N identical coupled nodes can be obtained.

$$\begin{aligned} dx_i(t) &= \tilde{x}_i(t) - \mu x_i(t) + \sum_{j=1}^N a_{ij}(t) x_j(t) dt, \\ d\tilde{x}_i(t) &= \left[(\mu - \Phi)(\tilde{x}_i(t) - \mu x_i(t)) - \rho x_i(t) \right. \\ &\quad \left. + \sum_{j=1}^N a_{ij}(t) \tilde{x}_j(t) \right] dt + \gamma x_i(t) dB(t), i \in \mathfrak{N}, \end{aligned}$$

where $a_{ij}(t)$ are the weights of the edges between node i and node j . Meanwhile, we have

$$\begin{aligned} dX^i(t) &= f^i(X^i(t), t) + \sum_{j=1}^N a_{ij}(t) X_j(t) dt \\ &\quad + g^i(X^i(t), t) dB(t), i \in \mathfrak{N}, \end{aligned} \quad (14)$$

in which $X^i(t) = (x^i(t), \tilde{x}^i(t))^T$, $g^i(X^i(t), t) = (0, \gamma x^i(t))^T$,

$$f^i(X^i, t) = \begin{pmatrix} (\mu - \Phi)(\tilde{x}^i - \mu x^i) - \rho x^i \\ (\mu - \Phi)(\tilde{x}^i - \mu x^i) - \rho x^i \end{pmatrix}.$$

Then, set system (14) as the driving system, and consider the following controlled response system:

$$\begin{aligned} dY^i(t) &= f^i(Y^i(t), t) + \sum_{j=1}^N a^{ij}(t) Y_j(t) dt + U^i(t) dt \\ &\quad + g^i(Y^i(t), t) dB(t), i \in \mathfrak{N}, \end{aligned} \quad (15)$$

where $Y^i(t) = (y^i(t), \tilde{y}^i(t))^T$, $\tilde{y}^i(t) = \dot{y}^i(t) + \mu y^i(t)$, $U^i(t)$ is the control input of the i th node, which is characterized as follows:

$$U_i(t) = \begin{cases} b^i(X^i(\delta(t)) - Y^i(\delta(t))), & t \in [\omega T, \omega T + \theta), \\ 0, & t \in [\omega T + \theta, (\omega + 1)T), \end{cases} \quad (16)$$

where $\omega = 0, 1, \dots, i \in \mathfrak{N}$. Let $E^i(t) = Y^i(t) - X^i(t)$, then we can obtain the synchronization error system as follows:

$$\begin{aligned} dE^i(t) &= f^i(Y^i(t), t) - f^i(X^i(t), t) + U^i(t) + \sum_{j=1}^N a_{ij}(t) E_j(t) dt \\ &\quad + [g^i(Y^i(t), t) - g^i(X^i(t), t)] dB(t), i \in \mathfrak{N}. \end{aligned} \quad (17)$$

5.1. Synchronization analysis for TSCOs on strongly connected digraph

In this subsection, we mainly study the synchronization of TSCOs on strongly connected digraph, and a synchronization criterion will be given. We first make some notations for the convenience of the research. Set

$$\begin{aligned} b &= \max_{i \in \mathfrak{N}} \{b^i\}, \bar{c} = \max_{i \in \mathfrak{N}} \{\bar{c}^i\}, c = \min_{i \in \mathfrak{N}} \{c^i\}, c = \max_{i \in \mathfrak{N}} \{c^i\}, \\ \bar{\delta} &= \max_{i \in \mathfrak{N}} \left\{ 2\xi + \gamma^2 + 2Na + \frac{c_i}{c} \right\}, \\ \xi &= \max_{i \in \mathfrak{N}} \left\{ \mu^2 + 4(\mu - \Phi)^2 \mu^2 + 2\rho^2 + 4(\mu - \Phi)^2 \right\}, \\ \beta &= \min_{i \in \mathfrak{N}} \left\{ 2b^i - 2\xi - \gamma^2 - 2\lambda_1 - 2Na - \frac{k(\tau)}{1 - k(\tau)} \frac{b^2}{2\lambda_1} \right\} - \frac{\bar{c}}{c}, \end{aligned}$$

in which λ_1 is a positive constant, and

$$k(\tau) = 2 \left(6\tau^2 \bar{c} b^2 + \frac{2\bar{c}^2}{c^2} \tau^3 \xi^2 + 8N^2 a^2 + 3\gamma^2 \right) \exp \left(\frac{2\bar{c}}{c} \tau^3 \xi^2 + 8N^2 a^2 + 3\gamma^2 \right).$$

Then, the following theorem can be derived.

Theorem 3. If Assumption 1 holds, and there exists a sufficiently small τ such that

$$k(\tau) < 1, \beta > 0, \theta\beta - (T - \theta)\delta > 0, \quad (18)$$

then, systems (14) and (15) achieve exponential synchronization under controller (16).

Proof. Consider that

$$\begin{aligned} \|f^i(y^i, t) - f^i(x^i, t)\|^2 &\leq 2 + 4(\mu - \Phi)^2 \bar{c}^2 \\ &\quad + (\mu^2 + 4(\mu - \Phi)^2 \mu^2 + 2\rho^2) e^2. \end{aligned} \quad (19)$$

Besides, we can deduce that

$$\|f^i(y^i, t) - f^i(x^i, t)\| \leq \xi |E^i|. \quad (20)$$

$$\|g^i(y^i, t) - g^i(x^i, t)\| \leq \gamma |E^i|.$$

Hence, according to (19) and (20), Assumption 1 holds. In view of the condition (18), we know that all conditions in Theorem 1 are satisfied. Consequently, according to Theorem 1, systems (14) and (15) achieve exponential synchronization. The proof is completed. $\square$

5.2. Synchronization analysis for TSCOs on digraph without strong connectivity

In the above subsection, synchronization of TSCOs on strongly connected digraph was researched. In this subsection, we will turn our focus to synchronization of TSCOs on digraph without strong connectivity.

For the convenience, we make some notations. Set

$$\begin{aligned} \hat{a} &= \max_{i,j \in \mathfrak{N}} \{a^{ij}\}, b = \max_{i \in \mathfrak{N}} \{b^i\}, \bar{\epsilon} = 2\xi + \gamma^2 + 4Na, \\ \epsilon &= -2\xi - 2b + 2\lambda_1 + \gamma^2 + 4Na + \frac{b^2}{2\lambda_1} \frac{k(\tau)}{1 - k(\tau)}, \end{aligned}$$

in which λ_1 is a positive constant, and

$$k^1(\tau) = 2 \left(6\tau^2 b^2 + 4\tau^3 \xi^2 + 4Na^2 + \gamma^2 \right) \exp \left(4\tau^3 \xi^2 + 4Na^2 + \gamma^2 \right).$$

Then the following theorem can be presented.

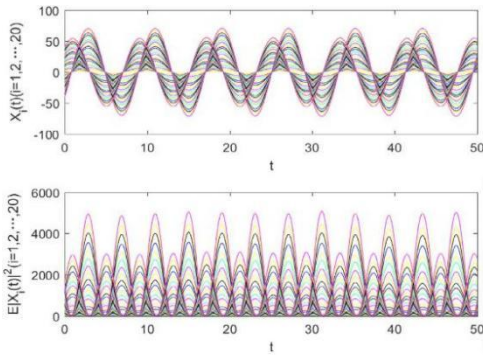


Fig. 4. The trajectory and mean square figures of drive system (14).

Theorem 4. When $i \in \cup^k D(T^{1k})$, all the conditions of Theorem 3 are satisfied. When $i \in \cap^k \cup^k D(T^{1k})$, if there exists a sufficiently small τ such that

$k_1(\tau) < 1, \epsilon > 0, \theta \epsilon - (T - \theta)\bar{\epsilon} > 0$,
then, systems (14) and (15) achieve exponential synchronization under controller (16).

Proof. According to (19) and (20) and all conditions in Theorem 4, we can deduce that Assumptions 1–3 hold, which implies that the conditions of theorem are satisfied. Consequently, systems (14) and (15) achieve exponential synchronization. The proof is completed. $\square$

Remark 8. Many scholars have great interest on the dynamical properties of coupled oscillators systems. For example, in Guo et al. (2018), Guo et al. investigated the inner synchronization of a class of coupled oscillators systems via periodically intermittent control. The stabilization of uncertain complex-valued coupled oscillators systems was studied through periodically intermittent control in Wu, Li and Li (2019). Nonetheless, different from the papers mentioned above, we utilize feedback control based on the discrete-time state observations in the control time. So the control energy can be reduced and the efficiency can be increased.

Remark 9. Recently, synchronization of coupled oscillators has been extensively studied. In Liu and Li (2018), outer synchronization for delayed coupled oscillators was discussed and the coupling structure was constant. In Guo et al. (2018) and Wang et al. (2019), the authors studied the synchronization of the stochastic coupled oscillators with a constant coupling structure on strongly connected digraph. Compared with the aforementioned papers, this paper studies synchronization of stochastic coupled oscillators without strong connectivity and with time-varying coupling structure via PIDOC, which is more general.

6. Numerical test

In this section, we mainly present a numerical example to illustrate the effectiveness and feasibility of our results.

Here, consider digraph $(\mathcal{G}, A(t))$ with 20 nodes (see Fig. 2), where $A(t) = (a^{ij}(t))_{20 \times 20}$ is the weight matrix of it. Besides, from Fig. 2, we can see that the network with 20 nodes is not strongly connected. By employing Tarjan's algorithm, we divide

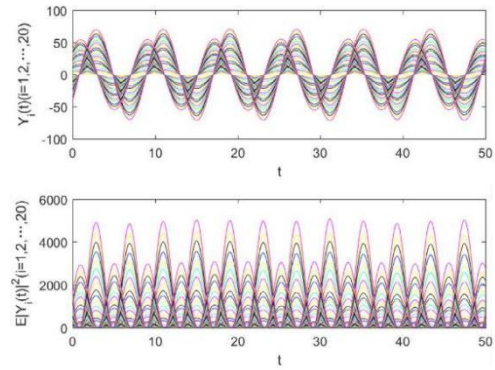


Fig. 5. The trajectory and mean square figures of response system (15).

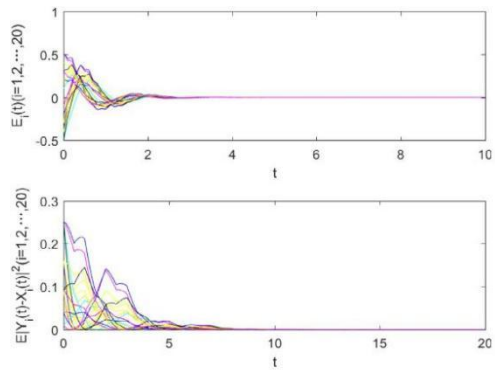


Fig. 6. The trajectory and mean square figures of error system (17).

it into three layers (see Fig. 3). Then we set some elements of $A(t)$ as follows, while other elements are defined as 0.

$$\begin{aligned} a_{14} &= a^{2,1} = a^{32} = a^{43} = 0.001(e^{-t} + 0.2); a^{57} = a^{65} = \\ a_{71} &= a^{76} = 0.001(0.7e^{-t} + 0.63); a^{89} = a^{95} = a^{98} = a^{10,12} = \\ &0.001(2e^{-t} + 1.53); a^{11,10} = a^{12,11} = a^{13,17} = a^{14,13} = 0.001(e^{-t} + 0.03); \\ a_{15,10} &= a^{15,14} = a^{16,2} = a^{16,15} = 0.001(1.3e^{-t} + 0.35); \\ a_{17,16} &= a^{18,20} = a^{19,18} = a^{20,13} = a^{20,19} = 0.001(0.1e^{-t} + 1). \end{aligned}$$

Now, choose some constants as $\mu = 0.1, \Phi = 0.003, \rho = 0.1, \gamma = 0.01, T = 1, \theta = 0.97, \lambda^1 = 2, \tau = 0.005$ and $b^i = 16, (i = 1, 2, \dots, 20)$. Then, we can obtain that

$$\delta = \delta^i = 0.01, \beta = 1.7913 > 0, \epsilon = 9.7914 > 0,$$

$$k(\tau) = 0.1061 < 1, k^1(\tau) = 0.0800 < 1,$$

$$\theta\beta - (T - \theta)\bar{\delta} = 1.4840 > 0, \theta\epsilon - (T - \theta)\bar{\epsilon} = 9.3080 > 0.$$

As a consequence, it is obvious that all conditions in Theorem 4 are satisfied. Hence, we can get the mean squares of drive system (14) and response system (15) in Figs. 4 and 5, respectively. Besides, the mean square of error system (17) are exhibited in Fig. 6. Meanwhile, the trajectory of control (16) is shown in Fig. 7. From those figures, we can derive that drive system (14) and response system (15) achieve exponential synchronization, which verifies the feasibility and effectiveness of our results.

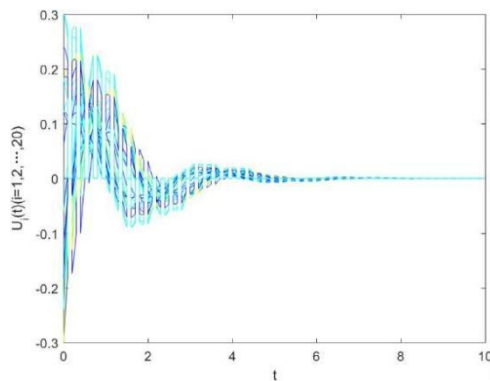


Fig. 7. The trajectory of control (16).

7. Conclusions

In this paper, the exponential synchronization of SCNWSCs was studied. A novel PIDOC was introduced to analyze the exponential synchronization of the general stochastic complex network with time-varying coupling structure. Furthermore, the PIDOC will degenerate into a regular FCBDSO when the control width tends to the control period. Besides, two corollaries about synchronization in mean square and asymptotical synchronization in mean square were obtained. In our future work, we will consider the following topics: 1. We will use periodically intermittent pinning discrete observation control to study the exponential synchronization of SCNWSCs, which can achieve synchronization of networks economically and effectively. 2. We will consider the case that the duration τ between two consecutive observations becomes time-varying $\tau(t)$.

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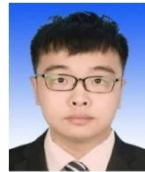
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